

Angel Reynoso

Final Project

CE2070

### ***Final Project: 2D Conservative Vector Solver***

**Problem:** In my Calculus III class we needed to use cross partial check on line integrals that calculate the work done by a vector field along a curve. This was easy but time consuming if the integral was big or had lots of variables. I wanted to create an app that was able to use the cross partial check on line integrals and if true would then solve for a potential function. Using the potential function you could use the fundamental theorem for line integrals.

#### **Application:**

The user is prompted to enter two components of the vector field,  $P(x,y)$  and  $Q(x,y)$ . Next, they must enter the starting points and the ending points of the vector field.

When the information needed is entered, they must click solve and the program will perform the cross partial check. This consists of taking the derivatives of  $P(x,y)$  with respect to  $y$  and  $Q(x,y)$  with respect to  $x$ . If the derivatives of the functions are equal, they pass the cross partial check test, and the program proceeds with the rest but if it fails the test they will be prompted with a text saying that the vector field is not conservative since it failed the test.

If the cross partial check passes, the program then finds a potential function by integrating  $P$  with respect to  $x$  and adding a constant of  $g(y)$ . Then deriving the answer with respect to  $y$  and getting  $g'(y)$ . Then setting it equal to  $Q$  and solving for  $g(y)$  and combining the functions with the first step. Before continuing to the Fundamental Theorem of Line Integrals the program verifies that the potential function is correct by taking the derivative and comparing it to the original vector field.

Finally, the program applies the Fundamental Theorem of Line Integrals by evaluating the potential function at the starting point and the end point given by the user. Then subtracting the end value by the start value( $f(\text{end}) - f(\text{beginning})$ ). This produces the value of the line integral without requiring the curve to be parameterized and makes the program even more efficient by not having to integrate the potential function.

**Summary:** The application checks the cross partials, finds and verifies a potential function, and uses the Fundamental Theorem of Line Integrals to calculate the final answer. It also displays each step so the user can follow the mathematical process. It follows the same procedure taught in Calculus III, reduces calculation errors, saves time on repetitive computations, and displays each step so users can better understand the solution process.

Link To Website: <https://2d-conservative-vector-solver.streamlit.app/>